COMP6258 Reproducibility Challenge: NegMerge: Sign-Consensual Weight Merging for Machine Unlearning

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Paper Information:

Openreview URL:	https://openreview.net/forum?id=ZbWXovStjD
Is code available?	https://github.com/naver-ai/negmerge

Plan:

What is the motivation of the paper?

Machine unlearning seeks to remove specific, targeted knowledge (such as copyrighted data or toxic behaviours) from a trained neural network without the massive computational expense and carbon footprint of retraining from scratch. Existing methods, like Task Arithmetic [1], require extensive hyperparameter validation to find a single optimal unlearning direction. NegMerge is motivated by the need to eliminate this computational waste by reusing all the fine-tuned candidate models generated during the validation phase rather than discarding them [2].

What are the key claimed contributions of the paper?

The paper's primary contribution is a novel aggregation method that merges multiple fine-tuned task vectors into a single, highly sparse update. NegMerge mathematically zeroes out up to 95% of the parameter shifts, by extracting a sign-consensus, which only retains weight updates that share the exact same directional sign across all models. The authors claim this sparsity allows the model to *forget* targeted data while suffering significantly less collateral damage on the retain set compared to standard Task Arithmetic baselines.

What experiments are we aiming to do?

We will structure our experiments into two phases: reproduction and extension, strictly bounded by our hardware limitations of one 24GB RTX 3090 and 100 Google Colab compute units.

Phase 1: Baseline Reproductions First, we will reproduce the standard classifier unlearning scenario using the ResNet-18 architecture on the CIFAR-10 dataset, targeting a 10% random subset as the forget set. To generate the necessary pool of 10 to 30 fine-tuned models for the NegMerge algorithm, we will train using the AdamW optimiser with varying learning rates and weight decays for 30 epochs per model. Second, we will replicate the Vision-Language zero-shot unlearning scenario. Using a pre-trained CLIP ViT-B/32 backbone, we will unlearn the "Cars" dataset while measuring general zero-shot retention on ImageNet. Here, model variations will be generated via RandAugment transformations, training for up to 70 epochs. To manage our VRAM constraints, we will implement dynamic accumulation of the sign-consensus mask in memory, sequentially loading checkpoints to maintain an $O(1)$ memory footprint.

Phase 2: Novel Extension for Sustainable LLM Unlearning The authors claim NegMerge scales to Large Language Models (LLMs). We will critically evaluate this by applying NegMerge to the Qwen2.5-7B-Instruct model using the TOFU (Task Of Fictitious Unlearning) dataset. Due to strict memory constraints, computing sign-consensus over 14GB of full-rank weight updates is impossible on our hardware. We will extend the methodology by applying it exclusively to QLoRA (Quantised Low-Rank Adaptation) matrices.

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The base model will be quantised at 4-bit precision, and we will train LoRA adapters (rank 8) on the TOFU forget splits. We will execute the NegMerge algorithm strictly over these low-rank matrices. For evaluation, we will use the TOFU metrics suite [3]. We will analyse ROUGE-L for model utility preservation and the Kolmogorov-Smirnov Forget Quality metric. The latter will verify if our unlearned model's Truth Ratio distributions match a gold-standard retain model, establishing if NegMerge provides a genuinely sustainable framework for editing modern LLMs.

References

[1] G. Ilharco, M. T. Ribeiro, M. Wortsman, S. Gururangan, L. Schmidt, H. Hajishirzi, and A. Farhadi, "Editing Models with Task Arithmetic," in *Proceedings of the 11th International Conference on Learning Representations*, 2023. [Online]. Available: https://arxiv.org/abs/2212.04089

[2] H. S. Kim, D. Han, and J. Choe, "NegMerge: Sign-Consensual Weight Merging for Machine Unlearning," in *Proceedings of the 42nd International Conference on Machine Learning, Vancouver, Canada*, 2024, version Number: 2. [Online]. Available: https://arxiv.org/abs/2410.05583

[3] P. Maini, Z. Feng, and A. Schwarzschild, "TOFU: A Task of Fictitious Unlearning for LLMs," Jan. 2024. [Online]. Available: https://arxiv.org/abs/2401.06121